

Request for Information on the Development of an Artificial Intelligence (AI) Action Plan

Agency:

Networking and Information Technology Research and
Development (NITRD) National Coordination Office (NCO),
National Science Foundation

Submitted by:

Thomas Kulik, General Counsel, MythWorx, LLC

This document is approved for public dissemination. The document contains no business-proprietary or confidential information. Document contents may be reused by the government in developing the AI Action Plan and associated documents without attribution.

Introduction

The key to America's dominance in AI is the development of AGI (Artificial General Intelligence) that goes far beyond typical LLMs such as OpenAI and DeepSeek in terms of capability and energy efficiency. If the Federal Government and U.S. Big Tech can achieve those dual goals, it will be very difficult for foreign adversaries to catch up.

While DeepSeek roiled the markets and the U.S. Big Tech and energy stocks with its more efficient and specialized "Mixture of Experts" methodology vs. OpenAI's more consistent "Dense Transformer" approach, AGI will be significantly more disruptive.

A Congressional Commission has recommended a "Manhattan Project" to develop AGI with public-private cooperation, but most AI leaders predict that AGI is still years into the future.

Meanwhile, a company in Texas named MythWorx has already developed a proven AGI capability. Its powerful AGI platform has been in development over the past 27 years and is just now emerging from stealth mode, having been working on multiple DoD contracts over the past two years.

Exploring the capabilities of a known AGI platform provides insight into how AGI can be leveraged to increase U.S. competitiveness.

Policy Refinement

AGI provides the strongest engine to unleash the power of free markets, world-class research institutions, and entrepreneurial spirit. Additionally, to maintain world AI leadership, the United States must “steal a march” to further develop AGI so neither ideological bias nor engineered social agendas has a chance to stunt its potential. With that in mind, we recommend the following policy to enhance action plans that gain and maintain America’s global AI dominance and to promote human flourishing, economic competitiveness, and national security.

Policy: The United States will promote and maintain global AGI dominance.

- a. **Action Plan Item:** Throughout 2025, apply the ARC-AGI version 2 benchmark leader results (see <https://arcprize.org/>) and aggressively resource top contenders to provision and deploy AGI’s power across our government, industry, and academia.
- b. **Action Plan Item:** Focus on the compute time and power requirements of the top ARC-AGI test performers to identify the most efficient AGI platforms to ensure scalability and resource efficiency. Real AGI will require far less compute power than current AI platforms.
- c. **Action Plan Item:** Focus on top ARC-AGI test performers that are capable of running on domestically-produced neuromorphic CPUs to reduce reliance on foreign GPU chip production.
- d. **Action Plan Item:** By January 2026, develop and execute further AGI benchmarks to expand the scope and recognition of AGI capability. Aggressively pursue and resource contenders to further provision and deploy AGI’s power across our government, industry, and academia.

Background

First, it's important to explore OpenAI vs. DeepSeek on how they approach their LLM strategy and why the market reacted the way it did to DeepSeek's introduction:

OpenAI

OpenAI's ChatGPT is like a librarian who has read all the books in the library before a question is asked and then gives an answer based on the vast quantity of books it has read, highlighted, and bookmarked. This approach requires more time and power before the question is asked, but uses less time and power while answering. Hence, OpenAI's time-consuming and expensive process requires an enormous amount of electricity-hungry Nvidia computer chips to read those books in advance.

DeepSeek

DeepSeek's R1 chatbot is like a team of expert librarians that haven't read all the books, but are trained to hunt out the right book within a specific field of expertise for the answer after a question is asked. This approach requires less time and power before the question is asked, but uses more time and power while answering. Hence, DeepSeek's shortcuts help it train AI at a fraction of the cost of OpenAI and competing models.

However, the introduction of AGI will be a significant leap forward in capability given the following difference:

AI

AI refers to task-specific systems that can perform intelligent functions like language processing, image recognition, or automation but lack general reasoning. AI operates within predefined rules and training data but does not possess human-like reasoning or adaptability beyond its trained scope.

AGI

AGI is a form (currently thought of by most as hypothetical) of AI with human-like cognitive abilities, capable of learning, reasoning, and adapting across any task without training. AGI, if achieved, would represent a true digital mind with self-awareness and general problem-solving skills.

The Strategic Nature of AGI

“

The Commission recommends:

Congress establish and fund a Manhattan Project-like program dedicated to racing to and acquiring an Artificial General Intelligence (AGI) capability. AGI is generally defined as systems that are as good as or better than human capabilities across all cognitive domains and would surpass the sharpest human minds at every task.

- U.S.-China Economic and Security Review Commission 2024 Annual Report to Congress, Part II, Technology and Consumer Product Opportunities and Risks, Chapter 3: U.S.-China Competition in Emerging Technologies¹

¹<https://www.uscc.gov/annual-report/2024-annual-report-congress>

The Test for Real AGI

The industry-defining test of AGI is the ARC-AGI test², which assesses AGI capability with hundreds of novel and unique puzzles designed to be easy for humans to solve, but difficult for AI to solve.

To fully grasp the significance of real AGI, one AGI platform, MythWorx, achieved a perfect 100% score on the ARC-AGI-1 test without any pre-training. By contrast, the second highest score is currently held by OpenAI's o3 chatbot at 87.5%. OpenAI's platform required pre-training on 75% of the puzzles and it's notable that the time and power requirements shown below did not include all of the pre-training effort, but only the test taking effort.

The potential of AGI (as demonstrated by MythWorx) compared to the current standard of AI in terms of pre-training requirements, compute time, and compute effort is highly significant and highlights the importance of the need for the U.S. to rapidly advance AGI capability.

	Typical AI (OpenAI o3)	AGI (MythWorx)
Score	87.5%	100%
Training	Pretrained	No pre-training
Compute time for test	23 hours	4 hours
Compute effort for test	9,500,000 watts	208 watts

²<https://arxiv.org/pdf/2412.04604>

The AGI Difference

Parameters and Impact on Compute Effort: AGI platforms such as MythWorx stand apart from traditional AI models by overcoming the inherent limitations of pre-trained models. Unlike traditional LLMs, MythWorx does not rely on static models, instead utilizing dynamic learning processes to enhance contextual understanding, emotional intelligence, and adaptability in human-machine interactions. MythWorx dynamically adjusts its parameters in real time, leading to substantial reductions in energy consumption. This efficiency not only makes MythWorx more scalable for various applications but also positions it as a sustainable AI solution in an era where computational cost and environmental impact are growing concerns.

Knowledge Graph and Emotional Intelligence: With a comprehensive and continuously updated knowledge graph, MythWorx delivers more flexible and natural responses than conventional AI models. This adaptability enables deeper user customization and higher levels of contextual awareness and emotional intelligence. A key differentiator for MythWorx is its superior contextual understanding and human-like responsiveness. The system's advanced knowledge graph allows for deeper comprehension of nuanced conversations, making interactions more fluid and personalized. Unlike conventional LLMs, which generate responses based on statistical likelihood, MythWorx actively interprets intent and emotion, enabling more natural and engaging exchanges.

No Labels and No Bias: Unlike LLMs that require extensive labeled datasets, MythWorx learns dynamically without the need for predefined labels, significantly reducing data preparation time. Traditional AI models are prone to inherent bias due to training on curated datasets, whereas MythWorx mitigates bias by continuously adapting to new information and evolving its responses. Additionally,

while LLMs often face functional constraints tied to predefined architectures, MythWorx operates without such limitations, ensuring greater flexibility and responsiveness.

Contextual Continuity, Persistence, and Capacity: MythWorx also outperforms LLMs in terms of token context and processing capacity. While conventional models are limited by a fixed token context window, MythWorx dynamically extends its memory, allowing for more comprehensive and coherent long-form interactions. The system's ability to recall past interactions in real time enhances its contextual continuity, making it particularly valuable in applications requiring persistent memory and adaptive learning. Unlike static LLM architectures, which demand significant storage for pre-trained weights and models, MythWorx employs a more efficient, storage-conscious approach that optimizes memory use without sacrificing performance.

Flexible Architecture and Increased Security: MythWorx is built with an advanced platform architecture that ensures seamless deployment across various environments. Whether implemented on cloud-based systems, edge computing devices, or secure enterprise environments, MythWorx adapts to different operational demands with minimal reconfiguration. This cross-platform compatibility, combined with its low computational overhead and real-time adaptability, positions MythWorx as a transformative AI solution for organizations seeking superior intelligence, efficiency, and scalability. Any AI must operate from a standpoint of "security first." MythWorx incorporates advanced cybersecurity capabilities by design to ensure that all interactions remain safely secured. Unlike other approaches, MythWorx integrates these capabilities from the operating system level through to the AI, creating an unsupervised, automated and continuously learning system to intelligently address the continuously evolving threat landscape.

Capability Comparison: The following table summarizes the capability difference between a typical AI platform such as OpenAI or DeekSeek and a true AGI platform like MythWorx.

	Typical AI	AGI (MythWorx)
Data Input	Manual	Automatic
Learning Style	Expensive training and periodic retraining	Low-cost continuous learning
Adaptability	Non-Adaptive	Adaptive, dynamically modifies connections
Style of learning	Inert repository	Active reasoning
Interactivity	Static	Dynamic
Timing	Episodic	Realtime
Level of Supervision	Supervised	Unsupervised
Labeling	Required	Not required
Potential Bias	Biased	Unbiased
Functionality Constraints	Bounded	Unbounded
Training Data Structuring	Structured	Unstructured
Token Context	Generative pre-trained transformers	Dynamic multi-modal graph-based
Processing Capacity	1 trillion parameters	100 quintillion parameters
Context Window	32,000 token limit	No token limit

	Typical AI	AGI (MythWorx)
Security	Highly vulnerable	Highly secure
Operational Level	Application level	Application and Operating System level
Efficiency	Inefficient	Highly efficient
Computation Effort	High computational overhead using power-hungry GPUs/TPUs for even simple queries	Asynchronous, event-driven computation that processes only necessary data in parallel
Memory and Storage Efficiency	High memory bandwidth due to constantly shuffling data between storage and processors	Low memory bandwidth due to using in-memory computing, can perform analog processing on neuromorphic chips
Platform Characteristics	Cloud dependent on large-scale datacenters	Edge computing ready with the ability to run on small, low-power devices

About MythWorx

The Launch: The MythWorx chief scientist is a brilliant savant and software engineer who wanted to replicate the functionality of the human brain, specifically in the style of how he learned. He began writing the code for what has grown into MythWorx in 1997, well before the term “AGI” came into use.

Growing the AGI Creation Team: The chief scientist was eventually joined by other engineers and as the AGI platform became more and more capable, it began to develop code on its own to expand its capability within our safety guardrails. The AGI platform has since birthed new, much more efficient versions and is now on its fourth generation. The team has since expanded its expertise, but most coding is done by the AGI platform itself.

Adding Advisors and Investors: The company added critical legal, financial, and leadership to the team over the past few years, but has continued to operate in stealth mode as it rapidly expanded its capability. The company is funded by the leadership team, which includes the GP of a global venture capital firm, angel investors, and a cleared Defense professional and AI investor. The firm has also been funded by revenue from past work with DoD.

DoD Projects: The firm has been deeply engaged with a large DoD contractor and is currently forming a JV with the firm to expand the relationship.

Commercial Availability: MythWorx is available for master license agreements or JVs with the U.S. Government, DoD, and U.S. Big Tech to ensure the U.S. can significantly extend and maintain its AI lead over the rest of the world.

AGI Has Arrived



www.MythWorx.ai